

Retail BI Architecture & Demand Forecasting

Power BI | Python | Data modelling | Forecast evaluation

BUSINESS CHALLENGE

How can a three-branch retailer compare performance across locations, products and customer groups, while using historical sales to support short-term decisions?

1,000

Sales records

\$325,721.75

Total sales

3

Retail branches

6

Product lines

Interactive Power BI dashboard



Executive dashboard: branch comparison, monthly trends, payments, product mix and KPIs.

What the analysis revealed

01

Branch Z led sales

It generated \$111,484.21 from 328 transactions; its average recorded sale was \$339.89.

02

Product performance varied

Food and beverages recorded the highest product-line sales at \$56,620.84.

03

Monthly totals need context

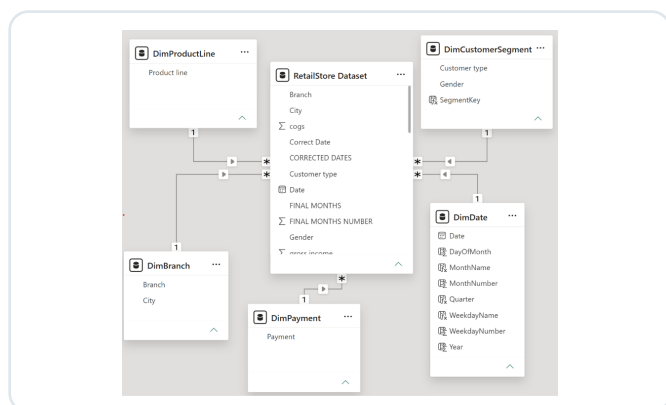
January: \$117,274.37; February: \$98,046.37; March: \$110,401.01. The shorter February needs daily-average comparison.

Data integrity: Mixed date formats required correction. The source field named "gross income" represented tax, not net profit; repeated invoice IDs were retained rather than automatically deleting sales.

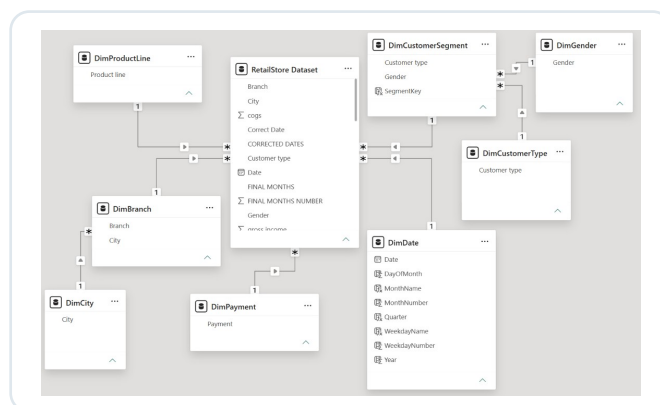
Design decisions & evaluated results

BI architecture & dimensional modelling

Designed a proposed flow from daily point-of-sale exports through staging, validation and a retail warehouse to Power BI reporting. Implemented Power BI star and snowflake model prototypes; the SQL warehouse, automated ETL and scheduled refresh were not deployed.



STAR SCHEMA

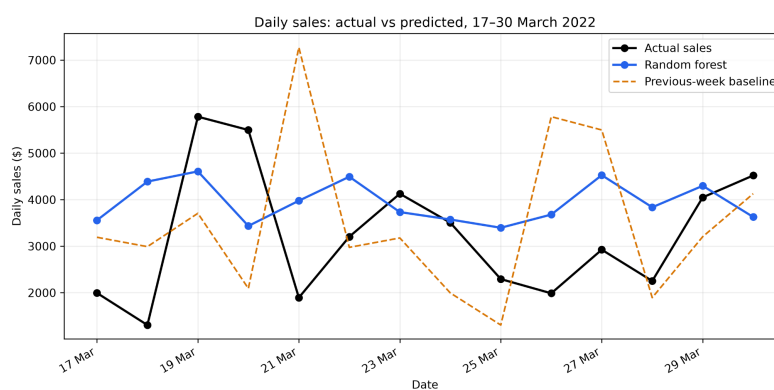


SNOWFLAKE SCHEMA

Design choice: The star schema has simpler reporting paths for this small dataset; the snowflake prototype normalises City, Gender and Customer Type into additional lookups. This is a design judgement, not a performance benchmark.

One-day-ahead sales forecast

A Random Forest model used day of week, prior-day sales, prior-week sales and a seven-day trailing average. The final 14 days (17-30 March 2022) were held out chronologically and compared with a previous-week baseline.



\$1,345.58

Random Forest MAE

\$1,813.09

Weekly baseline MAE

25.8%

Lower MAE vs baseline

Next-day estimate: \$3,589.16 for 31 March 2022 (not observed or verified). Limits: only 89 observed dates and a 14-day test; forecast errors remain substantial. Inventory, promotions and holiday signals were unavailable, so this is a planning prototype—not a validated stocking system.

Based on postgraduate coursework (2026); original dataset metrics shown.